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Cognitive Development



Perspective taking and cognitive flexibility in the Dimensional Change Card Sorting (DCCS) task

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ABSTRACT

In a study with 79 3-year-olds, we confirm earlier findings that separating the sorting dimensions improve children's performance on the Dimensional Change Card Sorting (DCCS) task. We also demonstrate that the central reason for this facilitation is that the two sorting dimensions are not integral features of a single object. Spatial separation of the sorting dimensions has no additional significant influence. This finding highlights the important role of objects with respect to children's attentional flexibility. Implications for current theories on the DCCS task and for the development of perspective taking and cognitive flexibility are discussed.

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Flexibility is a key competency in daily life, allowing us to adjust our behavior to the changing demands of the environment. Furthermore, cognitive flexibility is necessary for effective problem solving and creativity, and it allows us to use the regulatory strategy of reappraisal. Flexibility falls under the umbrella term of "executive function," which refers to higher cortical functions that enable conscious control of thought, action, and emotion. In addition to flexibility, executive functions are critical for planning, interference control, regulation of attention, and inhibition of inappropriate actions.

The development of flexibility follows a U-shaped function, with flexible shifting improving during childhood and declining with advancing age (Cepeda, Kramer, & Gonzalez de Sather, 2001). A significant developmental change in flexibility occurs between 3 and 5 years of age. This marked improvement in children's flexible behavior has often been demonstrated using the Dimensional Change Card Sorting (DCCS) task (Zelazo, Frye, & Rapus, 1996; Zelazo, Müller, Frye, & Marcovitch, 2003; see Zelazo, 2006, for a detailed protocol). In the DCCS task, bivalent test cards showing pictures that vary on two dimensions (e.g., red pears and blue apples) must be sorted first according to one

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dimension (e.g., color) and then according to the other dimension (e.g., shape) into two boxes, each marked with a target card (e.g., a red apple and a blue pear). Each test card matches one target card on one dimension and the other target card on the other dimension. In the so-called pre-switch phase, children are told a rule, for example, the color rule: “Sort all the blue ones into the box portraying something blue and sort all the red ones into the box displaying something red.” Typically, 3-year-olds have no problem doing so. However, they usually have severe difficulties when, after a series of trials, the sorting rule changes. Now, in the so-called post-switch phase, the child is instructed to sort the cards according to a new dimension, for example shape. It is not until about 4 years of age that children are able to switch flexibly to sorting on the post-switch dimension. Most 3-year-olds (especially younger 3-year-olds) perseverate and continue to sort according to the first dimension (Zelazo et al., 1996).

When speaking of *flexibility*, we often fail to distinguish two different kinds, *response flexibility* and *representational flexibility* (sometimes also referred to as *attentional flexibility*; see also Zelazo, Reznick, & Spinazzola, 1998, who distinguish between representational flexibility and response control in a multistep multilocation search task). In order to show flexible behavior, a person often needs both to think flexibly (e.g., by switching representations or cognitive sets) and to act flexibly (e.g., by switching responses). Although early accounts of children’s performance on the DCCS task (Carlson & Moses, 2001; Perner, Stummer, & Lang, 1999) explained children’s difficulty as a failure to inhibit interfering response tendencies, a host of subsequent research on the DCCS task showed that children’s perseverative sorting behavior is due not to inflexibility at the response level, but to problems with attention shifting. For example, Jacques, Zelazo, Kirkham, and Semcesen (1999) showed that 3-year-olds also have problems in the DCCS task when they do not respond by sorting themselves but by judging whether a puppet is sorting the cards correctly: In the post-switch phase, children judged the puppet to be correct when the puppet perseverated; when the puppet sorted correctly, they judged the puppet to be wrong. Moreover, 3-year-olds perseverate even after a single pre-switch trial, when one would expect the pre-switch-response to be less well-established (Zelazo et al., 1996, Exp. 2), or when a verbal response instead of a manual response is required (Zelazo et al., 1996, Exp. 3). In addition, Towse, Redbond, Houston-Price, and Cook (2000, Exp. 1; see also Rennie, Bull, & Diamond, 2004, for similar results) showed that new response options (two additional sorting trays) in the post-switch phase do not significantly influence children’s response pattern.

Furthermore, it has repeatedly been shown that 3-year-olds are well able to switch their response in card-sorting tasks that require a *reversal* of the response set in the absence of a dimension switch (Brooks, Hanauer, Padowska, & Rosman, 2003, Exp. 1; Kloo, Perner, Kerschhuber, Dabernig, & Aichhorn, 2008; Perner & Lang, 2002). In these reversal shift tasks, children had to switch from a “normal” shape game to a “silly” shape game. For example, in the “normal” shape game, the apples went to the apple target and the pears went to the pear target, but in the “reversed” or “silly” shape game, the apples went to the pear target and the pears went to the apple target. Also, Carlson (2005) found that most 3-year-olds – but not 2-year-olds – pass a reverse categorization task, in which they have to reverse a sorting rule. In sum, this evidence indicates that 3-year-olds are quite able to flexibly switch from one response to another, but they have difficulty in flexibly switching their attentional set from one dimension to another dimension.

Despite their profound difficulties in shifting between different aspects of the same object, 3-year-olds are able to shift to another sorting dimension if the two dimensions are not integrated within one object. For example, Kloo and Perner (2005) separated the two sorting dimensions by replacing the usual cards that show objects of a certain color (say, a blue apple) with cards on which the relevant object (apple) and the relevant color (blue) were realized as separate entities (an outline of an apple on one section of the card and a blue filled circle on another section). Diamond, Carlson, and Beck (2005) also used a separated version of the DCCS task, but in their study the spatial segregation of the two dimensions was less pronounced. They presented children with black shapes on colored backgrounds; that is, shape was a property of the foreground object, and color was a property of the background. Nevertheless, in both studies using these separated versions, children had much less difficulty. It should be mentioned that, mostly, in these studies dimensions were separated both on test and target cards, because Zelazo et al. (2003, Exp. 6) found no enhancing effect when dimensions were separated only on the test card (see Kloo & Perner, 2005, p. 49, for a discussion of these discrepant findings).

Unfortunately, in these separated versions of the DCCS task (Diamond et al., 2005; Kloo & Perner, 2005), two factors were confounded. In contrast to the standard DCCS, in the separated version (a) color and shape were not integrated within *one object*, and (b) color and shape were not displayed at the *same location*. Therefore, it is not clear whether the separated versions were easy (a) because switching between dimensions is easier when the two dimensions are not integrated within one object, or (b) because switching between dimensions is easier when the two dimensions are displayed at different locations.

It is important to know whether spatial or object-wise separation is crucial for the enhancing effect of these separated versions for young children, because this knowledge would help us to refine current theories on the development of cognitive flexibility. Furthermore, a systematic investigation of the effects of spatial- and object-wise separation is warranted because a host of research with adults has shown that space-based and object-based attentional mechanisms can be distinguished (and also interact) at a behavioral and at a neural level (Fink, Dolan, Halligan, Marshall, & Frith, 1997; Müller & Kleinschmidt, 2003; Mozer & Vecera, 2005). Therefore, knowledge about the effects of *spatial- vs.* object-wise separation could help to develop a unified account of the development of selective attention and attention switching as suggested by Hanania and Smith (in press).

Five major theories seek to explain children's difficulty with the DCCS task. Two of these accounts focus on difficulties at a conceptual level; the other three accounts focus on problems at the level of executive functions, such as attentional engagement/disengagement or working memory. Let us first consider the two conceptual accounts:

The traditional account of children's difficulty with the DCCS task is the *Cognitive Complexity and Control* theory (CCC theory; Zelazo & Frye, 1997; Zelazo et al., 2003). The CCC theory posits that the DCCS task requires a higher-order rule that integrates two conflicting pairs of rules. According to CCC theory, although 3-year-olds are able to represent the pre- and post-switch pairs of rules, they fail to reflect on these two contradicting rule pairs. Therefore, they cannot decide to switch to the post-switch rules. That is, 3-year-olds' inflexibility is linked to developmental constraints on the representation and use of a higher-order rule. According to CCC theory, an ability to self-reflect is necessary for switching on the DCCS task.

However, a number of studies indicate that 3-year-old children's perseverative performance cannot be explained exclusively by rule complexity, as defined by CCC theory, because different research groups have shown that the DCCS task can be modified in ways that preserve the embedded rule structure but markedly improves 3-year-olds' performance. These include use of congruent flankers (Jordan & Morton, 2008); highlighting the switch (Mack, 2007); use of target puppets (Perner & Lang, 2002); labeling (Kirkham, Cruess, & Diamond, 2003; Towse et al., 2000; but see Müller, Zelazo, Lurye, & Liebermann, 2008, for a failure to replicate the labeling effect).

Taking into account empirical evidence against the original CCC theory, Zelazo et al. (2003) proposed a revised version of CCC theory, the *Cognitive Complexity and Control-revised* (CCC-r) theory, which also includes the activation and inhibition of rules and the importance of conflict. With respect to the separation of dimensions, CCC-r theory may predict that separated versions are easier, because the visual separation of dimensions reduces complexity or conflict, or because the visual separation affects the dynamic balance of attention to the two dimensions. However, it should be noted that, at first, the proponents of CCC-r theory explicitly predicted that 3-year-olds should persevere in separated versions of the DCCS task "because it requires them to cross major branches of a tree-like [rule] structure." (Zelazo et al., 2003, p. 57); indeed, Zelazo et al. found that the separation of dimensions had no effect (Zelazo et al., 2003, Exp. 6). However, as discussed by Kloo and Perner (2005, p. 49), some methodological differences might explain the contrasting results.

The other account that focuses on a conceptual level is the *re-description* hypothesis proposed by Kloo and Perner (2003, 2005). They argued that 3-year-olds have problems on the DCCS task because they do not understand that things can be described differently from different perspectives. That is, in the standard DCCS, children must understand that, for example, a red apple can be identified as "a red thing" (in the color game) and as "an apple" (in the fruit game). However, if dimensions are separated (and attached to different objects), children need not understand that one object can be described in two different ways; they merely have to switch between objects, for example, switch

from sorting the colorless line drawings in the pre-switch phase to sorting the colored circles in the post-switch phase. According to the re-description account, the separation of the two dimensions into two different objects should be critical, but spatial separation of the two dimensions should not be important.

Apart from these two conceptual accounts, there are three other accounts of children's difficulties with the DCCS task that focus on executive control processes. One account that has received considerable empirical support suggests that *negative priming* contributes to children's difficulty (Müller, Dick, Gela, Overton, & Zelazo, 2006; see also Chevalier & Blaye, 2008). This account posits that the values of the irrelevant dimension (e.g., shape) have to be inhibited in the pre-switch (e.g., color) phase in order to sort correctly. In the post-switch phase, this inhibition has to be cancelled, when the formerly irrelevant dimension (i.e., shape) becomes relevant.

Research with adults has shown that physical separation of targets and distractors decreases interference and negative priming effects (Fox, 1994, 1995; Milham & Banich, 2005). For example, Fox (1994) showed that reducing the spatial separation between targets and distractors increases the magnitude of negative priming. Furthermore, in addition to the spatial proximity of targets and distractors, negative priming is also affected by object-based perceptual grouping (Fuentes, Humphreys, Agis, Carmona, & Catena, 1998). Therefore, according to the negative priming account, separated versions of the DCCS task should be easier than integrated versions. Furthermore, as spatial- and object-wise separation have been shown to affect the magnitude of negative priming (at least in adults), spatial- and object-wise separation should exert additive effects.

In contrast to the negative priming account, that focuses on problems with attentional disinhibition, Diamond and colleagues (Diamond et al., 2005; Kirkham et al., 2003) proposed an account focusing on problems with inhibition of attention, the "*attentional inertia*" account. According to this account, "... 3-year-old children's difficulty lies in disengaging from a mindset (a way of thinking about the stimuli) that is no longer relevant." (Kirkham et al., 2003, p. 451). Diamond and colleagues argue that the separated versions are easy because separating the two dimensions reduces the need for inhibitory control; the separated versions never require "inhibition of the initial way of thinking about the *objects* [*italics added*]." (Diamond et al., 2005, p. 710) That is, according to Diamond et al., object-wise separation of the two dimensions should be critical, whereas spatial separation should not play a major role.

Recently, several authors have pointed to the importance of *goal maintenance* for successful performance on the DCCS task (Marcovitch, Boseovski, & Knapp, 2007; Towse, Lewis, & Knowles, 2007). Problems with goal maintenance can become manifest as "goal neglect": disregard of a task requirement even though it has been understood and remembered (Duncan, Emslie, Williams, Johnson, & Freer, 1996). Marcovitch et al. (2007) showed that goal maintenance in preschool children can easily be disrupted using a modified version of the DCCS task. In addition to the standard conflict test cards (that have to be sorted differently depending on the rules), children also received redundant test cards (that have to be sorted the same regardless of the rules). Four- to 5-year-old children performed more poorly in the post-switch phase when a version with 80% redundant test cards was used than when a version with only 20% redundant test cards was used, possibly because redundant test cards do not require consistent activation of the rules (or the goal state) and therefore impair goal maintenance. Furthermore, Towse et al. (2007) reported an association between a child-appropriate goal neglect task (the selective image naming task) and the DCCS task. In its focus on working memory and the strength of goal representations, this account also has some similarity to the neural network model focusing on active *vs.* latent representations proposed by Morton and Munakata (2002). It is not clear how these goal maintenance accounts would explain the enhancing effect of the separated DCCS versions. However, proponents of goal maintenance accounts may argue that the separated versions are easier because separation makes the two different goals (color/shape) explicit on the cards.

To investigate the issue of *spatial- vs.* object-wise separation, in the present study we compared children's performance on three different card sorting tasks: the standard DCCS, a separated card sorting task with dimensions displayed at different locations (*spatially distinct version*), and a separated card sorting task with dimensions displayed at the same location (*overlapping version*). In the separated versions, dimensions were separated both on test and target cards.

1. Method

1.1. Participants

A total of 80 children were recruited from 12 different nursery schools. One child was excluded because of refusal. The final sample comprised 79 children (35 girls and 44 boys). Most participants were Caucasian and came from a middle-class background, although data about race and socioeconomic status were not systematically collected. Children's ages ranged from 3;0 to 3;11 years (mean age = 3;5 years, $SD = 2.81$ months).

To examine age trends, we divided children into three age groups: 19 children ranging in age from 3;0 to 3;2 (mean age 3;1 years, $SD = 0.82$ months), 31 children ranging in age from 3;3 to 3;5 (mean age 3;4 years, $SD = 0.77$ months), and 29 children ranging in age from 3;6 to 3;11 (mean age 3;8 years, $SD = 1.54$ months).

1.2. Design

Children were tested individually in one session of about 15 min. Each child received two card-sorting tasks, a standard DCCS and a separated DCCS. Children were randomly assigned to one of two separated conditions. Thirty-nine children were given a separated DCCS with dimensions displayed at different locations (*spatially distinct version*), and the other 40 received a separated DCCS with dimensions displayed at the same location (*overlapping version*). The two groups did not differ significantly in age (spatially distinct: $M = 40.74$ months, $SD = 2.81$ months; overlapping: $M = 40.78$ months, $SD = 2.83$ months; $t[77] = 0.049$, $p = .96$). The order of the two tasks, the material (clothes or fruit) used for each task, and the direction of shift (both tasks per child from color to shape or both tasks per child from shape to color) were counterbalanced. After the session, children received a small present for "doing so well."

1.3. Materials

Each of six sets of 12×7 cm cards consisted of two target cards and 12 test cards. In all versions, the two target cards were each affixed to $22 \times 14 \times 16$ cm boxes. The test cards could be placed into one of these boxes through a slit on the front side. Each task involved three phases: two demonstration trials, a pre-switch, and a post-switch phase.

We used two different color-form combinations: (a) blue and yellow trousers and shoes; (b) red and green apples and bananas. For each color-form combination, three different display modes were used (integrated, spatially distinct, and overlapping) resulting in six different sets of cards. In the two separated versions, dimensions were separated on test and target cards. Fig. 1 depicts the three different display modes of green and apple.

1.4. Procedure

The procedure was the same in all versions and followed the standard DCCS task (Zelazo, 2006). For example, in the fruit version, the experimenter pointed at the two target cards and explained the two dimensions (shape and color). Then, while pointing at the appropriate boxes, she said, "Now we are playing a game, the color game. In this game, all the red ones go to the red one. And all the green ones go to the green one." The experimenter sorted one test card (one red and one green) into each box. After these two demonstration trials, the pre-switch phase began, and children were asked to sort five cards on their own. On each trial, the experimenter randomly selected a test card, labeled the card by the relevant dimension only (e.g., "Here is a red one.") and asked the child to sort the card into one of the boxes ("Where does this card go in the color game?"). On each trial, children were told whether they had sorted the card correctly. If a card had been sorted incorrectly, the pre-switch rules were repeated.¹

¹ The use of feedback in the pre-switch phase differs from the standard DCCS task protocol as described by Zelazo (2006). Although in recent years, mostly no feedback is given in the pre-switch phase (Diamond et al., 2005; Kirkham et al., 2003), in

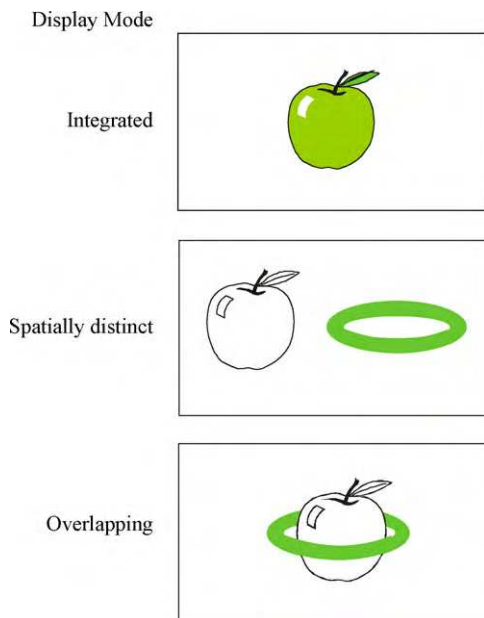


Fig. 1. Exemplary integrated, spatially distinct, and overlapping cards.

When the children had completed five trials, the rules changed, and the post-switch phase began. Children were told, “Okay, now we are going to play a NEW game, the fruit game. The fruit game is different: All the bananas go to the banana. And all the apples go to the apple.” Again, children had to sort five cards according to the new rules. They were not told whether a card had been placed correctly. However, as in the pre-switch phase, every time a card had been sorted incorrectly, the experimenter repeated the post-switch rules.

2. Results

In both the pre- and post-switch phase of each task, children were given a score between 0 and 5 based on the number of cards sorted correctly. During the pre-switch phase, performance was almost perfect. In the integrated version, one child sorted one card incorrectly; two children made one error in the spatially distinct version.

The main variable of interest was the number of correct responses in the post-switch phase. In all versions of the task, most children sorted at least four of five or none of the five post-switch cards correctly: 78.5% of the children in the standard DCCS, 84.6% of the children in the spatially distinct version, and 82.5% of the children in the overlapping version did so. Table 1 provides a summary of performance in the post-switch phases for each of the task versions.

Post-switch scores on the two versions each child performed were not normally distributed (Kolmogorov–Smirnov test). Also, due to the bimodal nature of the data, common data transformations did not result in normality. Therefore, children’s performance was examined as a dichotomous (“pass” or “fail”) variable. Children were classified as passing the post-switch phase if they sorted four or five of the five items correctly.

No significant differences were found by task order or dimension order. McNemar’s χ^2 tests (with continuity correction) confirmed that children performed better on the separated versions than on

their early work Zelazo and colleagues (Zelazo et al., 1996, Exp. 1; see also Perner & Lang, 2002; Perner et al., 2002) sometimes gave feedback in the pre-switch phase. In our study, we adhered to this earlier method, although it might be preferable to give no feedback in the pre-switch phase in order to make the two phases of the DCCS task more comparable.

Table 1
Number of post-switch cards sorted correctly on each card sorting task.

Task	Number of post-switch cards sorted correctly								M	SD
	N/n	0	1	2	3	4	5			
Standard DCCS	79	34	6	7	4	7	21	2.09	2.15	
Separated DCCS	79	7	4	4	5	20	39	3.82	1.62	
Spatially distinct	39	3	2	1	3	7	23	4.00	1.59	
Overlapping	40	4	2	3	2	13	16	3.65	1.66	

Note: Maximum score = 5.

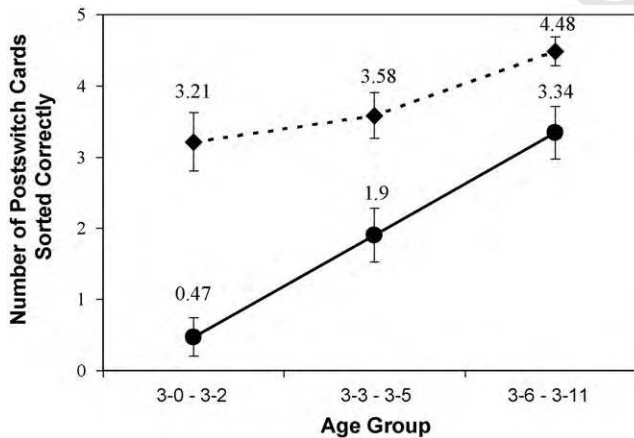


Fig. 2. Post-switch performance on the standard (solid line) and separated (dashed line) versions, by age group.

the standard DCCS, $\chi^2(1, N=79)=29.03, p<.001$ for the whole sample and for those receiving each version; $\chi^2(1, N=39)=15.06, p<.001$ for the subgroup receiving the spatially separated version; $\chi^2(1, N=40)=12.07, p<.001$ for the subgroup receiving the overlapping version. In sum, 31 children passed the separated version and failed the standard DCCS, but no child showed the reverse pattern. Children's performance on the two separated versions did not differ, $\chi^2(1, N=79)=.204, p=.65$; 76.9% passed the spatially distinct version, and 72.5% passed the overlapping version.

Performance on the standard DCCS improved significantly with age, $\chi^2(2, N=79)=14.51, p=.001$, with 5.3%, 32.3%, and 58.6% successfully switching in the three age groups respectively. In contrast, on the separated versions, performance in the three age groups did not differ significantly, $\chi^2(2, N=79)=3.60, p=.16$, with 63.2%, 71.0%, and 86.2% passing in the three age groups respectively. Separate McNemar's χ^2 tests in the three different age groups showed that children performed significantly better on the separated version than on the standard version in all age groups, all p 's $>.05$. Fig. 2 shows the developmental trend for the standard and separated versions.

3. Discussion

This experiment corroborates and extends the results of Diamond et al. (2005) and Kloo and Perner (2005) by highlighting the importance of objects. We confirm the finding that 3-year-old children perform significantly better on a DCCS task when dimensions are separated than when dimensions are integrated within one object. In addition, this study provides evidence that the central reason for the facilitation by the separated versions is that the two sorting dimensions are not integral features of a single object. The spatial separation of the two dimensions had no additional significant influence, suggesting that basic space-based attentional processes are not a crucial source of children's difficulties on the DCCS task.

Thus, a crucial factor in children's difficulties on the DCCS task is the integration of both sorting dimensions within one object. Set-shifting between different objects (e.g., shifting from sorting colored circles to sorting colorless line drawings) is easy, but set-shifting within one object (e.g., shifting from sorting a green apple according to color to sorting it according to shape) is difficult for 3-year-olds. The central difficulty for 3-year-old children seems to lie in shifting between different ways of thinking about one and the same object.

We first discuss this finding with respect to current theories regarding the DCCS task and then discuss implications for the development of perspective taking and attentional flexibility. Although this study clearly indicates that object-wise separation of the two sorting dimensions is essential for the ease of the separated versions, current accounts of preschoolers' difficulty on the DCCS task, apart from the attentional inertia hypothesis and the re-description hypothesis, do not explicitly specify that or why object-wise separation, but not spatial separation, should be critical.

The present results indicate that 3-year-old children's perseverative performance cannot be explained exclusively by rule complexity, as defined by the original CCC theory. The revised CCC-r theory, however, might argue that separation, for example, reduces the amount of conflict and, therefore, improves performance. But then CCC-r theory would also have to explicitly specify why object-wise separation is critical.

The negative priming account could explain the finding that separated versions of the DCCS task are easier, because research with adults has shown that spatial and object-wise separation of targets and distractors decreases negative priming effects (Fox, 1994; Fuentes et al., 1998). Therefore, the separated versions of the DCCS task may be easier for children because the separation reduces the negative priming effect. However, it is unclear how the negative priming hypothesis could account for the fact that in our study object-wise separation was most important for the ease of the separated task, whereas spatial separation did not exert a significant additional influence, as the spatial separation should further decrease interference and negative priming.

The present results may also be explained within a goal maintenance framework by arguing that the physical separation of the two sorting dimensions makes the goals of the task more explicit. However, this suggestion would have to be strengthened by empirical findings. For example, children's performance on the standard integrated DCCS task should be more strongly related to goal maintenance than performance on separated versions of the DCCS task.

In sum, the present findings do not rule out existing accounts of children's difficulty with the DCCS task. Rule following, negative priming, and goal maintenance may all play a role in children's performance on the DCCS task. But these accounts need to specify why object-wise separation/integration of the two sorting dimensions is so essential.

The attentional inertia and the re-description accounts explicitly emphasize the importance of objects. The attentional inertia account (Diamond et al., 2005) attributes children's difficulty to problems in inhibiting the initial way of thinking about an object. And the re-description theory (Kloo & Perner, 2005) posits that 3-year-olds do not yet understand that things can be described differently from different perspectives.

Although both the attentional inertia and the re-description accounts assume that the central problem for 3-year-olds on the DCCS task is the demand to switch between different mind-sets or descriptions of one object, they differ in their explanation of how this specific kind of flexible set-shifting develops. Whereas Diamond and colleagues argue that flexible switching requires sufficiently developed inhibitory control, Kloo and Perner (2005) argue that switching between different mind-sets of one object requires a mature conceptual understanding that the same objects can be described or viewed differently. Thus, further research is needed to clarify the developmental origin of the ability to shift between different mind-sets of one object.

Apart from the DCCS task, our finding that 3-year-old children have particular difficulty shifting between different mind-sets with regard to an object has implications for cognitive development in general. There is independent evidence from discrimination learning and free classification tasks that younger children perceive multidimensional objects as undifferentiated wholes, whereas older children and adults perceive these objects as conjunctions of separable features or dimensions (Hanania & Smith, in press). For this reason, younger children may find it difficult to attend selectively to a single

feature or a single dimension in the DCCS when integrated stimuli are used. Consequently, spatially separating the two dimensions improves performance.

In fact, there is a wealth of evidence that young children before the age of about 4 years have difficulty understanding that one entity (e.g., an object or a certain situation) can be described differently from different perspectives. They have problems understanding that other people may have a false belief (a representation of the real world that differs from one's own representation; Perner, 1991), that appearances may differ from reality (Flavell, Flavell, & Green, 1983), or that different perspectives can be taken on the same thing. The idea that children's understanding of multiple perspectives is a crucial development attainment can be traced back to Piaget (1929/1969). In Piaget's view, young children do not know that their own point of view is subjective and that other people may hold other perspectives on the world. Parallel developmental progress in children's understanding of perspective and in their performance on the DCCS task seems to be more than coincidental. The DCCS task has repeatedly been shown to be specifically related to perspective taking tasks (Carlson & Moses, 2001; Frye, Zelazo, & Palfai, 1995; Kloo & Perner, 2003; Perner, Lang, & Kloo, 2002).

Although children have particular difficulty shifting between different mind-sets with regard to the same object, research on object-based attention in adults shows an opposite pattern of results. Several studies (Baylis & Driver, 1993; Duncan, 1984; Müller & O'Grady, 2000) have demonstrated that switching attention within one object is easier than switching attention across objects. For example, Duncan (1984) showed that two judgments regarding the same object were more accurate than two judgments that concerned two different objects. (See Kloo, Perner, Trendl, Schmidhuber, & Aichhorn, 2008, for a similar finding on an adult's version of the DCCS task.) This suggests that attention switching regarding one object operates differently in children and adults. However, the exact nature of these differences remains to be determined.

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